Training Setup We trained all of the models using the unsupervised training procedure described in Section 3.2. For each dataset, we divided the auxiliary knowledge base into equal chunks of size 256 by concatenating or splitting the original chunks based on their length. We also added two special tokens, <BOS> and <EOS>, to demarcate the original chunks' beginnings and ends to preserve the documents' structure.

The models were trained using learning rates between 1×10^{-6} and 5×10^{-5}, which were found through a hyperparameter search. All models were trained on 4 NVIDIA A-100 GPUs for a maximum of 5 epochs and a batch size of 64.

Evaluation method All evaluations were done by appending each of the multiple-choice options to the question, followed by passing the concatenation through the model to get a log probability score per option. The highest score was interpreted as the model's choice and used for accuracy calculation. More formally, this means that in Equation (1) we say that $\mathcal{M}(q_n) = c_n$ if:

$$c_n = \arg\max_l \{\mathcal{M}(q_n \| a_n^1), \ldots, \mathcal{M}(q_n \| a_n^L)\}, \quad (4)$$

where $\mathcal{M}(q_n \| a_n^l) = \log P_{\mathcal{M}}(q_n \| a_n^l)$.

MMLU Results For each task and model, we compared four approaches: using just the base model, RAG, FT, and finally combining FT and RAG by using the fine-tuned model as the generator. Furthermore, we tested the MMLU tasks using both 0-shot and 5-shot scenarios. The full results are shown in Table 1. An aggregation of the relative accuracy gain, i.e.,

$$(\mathcal{L}_{\mathcal{M}',\mathcal{Q}} - \mathcal{L}_{\mathcal{M},\mathcal{Q}})/\mathcal{L}_{\mathcal{M},\mathcal{Q}}, \quad (5)$$

where $\mathcal{M}$ is the base model and $\mathcal{M}'$ is the knowledge-injected model, is shown in Figure 2.

In all cases, RAG performed significantly better compared to the base models. Furthermore, using RAG with the base model as the generator was consistently better than only fine-tuning. In some cases, using the fine-tuned model instead of the base model as the generator in the RAG pipeline improved results even further. However, this is not consistent and thus demonstrates the inherent instability of fine-tuning. Additionally, we found that the 5-shot approach boosts the results by a small margin in most cases, with a similar trend being observed in all of the different approaches.

Current Events Results The evaluation on the current events task is shown in Table 2. RAG proves particularly effective due to the one-to-one correspondence between the questions and the auxiliary dataset (see Section 4.3). Fine-tuning is not competitive with RAG. However, fine-tuning with multiple paraphrases still provides a significant improvement over the baseline. We note that combining RAG with fine-tuning shows inferior performance compared to RAG alone.

It is worth noting that although the questions are based on information the models were not exposed to during training, the results of the base models surpass $\frac{1}{L} = 0.25$. This can partially be explained by the models using reasoning and/or pre-existing knowledge when answering questions that are not independent of the past information. Some examples of this can be found in Appendix C.

Fine-Tuning vs. RAG: In the results of both the MMLU and current events tasks, a significant advantage for RAG over fine-tuning is evident. While fine-tuning improved results compared to the base model in most cases, it was not competitive with the RAG approach.

Several factors might contribute to this behavior. Firstly, RAG not only adds knowledge to a model but also incorporates context relevant to the question, a feature lacking in fine-tuning. Additionally, fine-tuning may impact other capabilities of the model due to a degree of catastrophic forgetting. Finally, it's plausible that unsupervised fine-tuned models might benefit from further alignment through supervised or RL-based fine-tuning, as evidenced by the vastly improved performance of Orca2 over the base Llama2.

6. The Importance of Repetition

Unlike the other tasks, where the model has been exposed to aspects related to the topic during pretraining, *current events* includes new information. In this case, standard regular fine-tuning not only did not improve the performance of Llama2 but also significantly degraded it. To improve the fine-tuning results, we explored augmentation of the data using paraphrases.

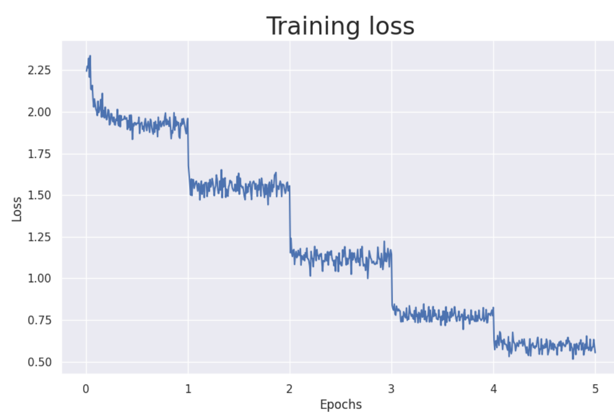

Figure 3. Training loss over time for Mistral-7B.

Data Augmentation Data augmentation is a well-established method for enhancing the performance of language models and has been surveyed extensively (Shorten

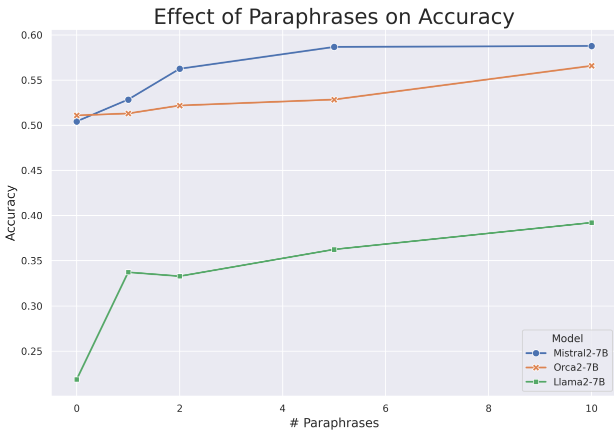

Figure 4. Model accuracy on the *current events* task as a function of the number of paraphrases.

et al., 2021). Using generative models for augmentations has also been used successfully to improve classification models in the past (Sharma et al., 2022). An example of data augmentation using paraphrasing can be found in Appendix B.

Monotonic Improvement This approach resulted in notable improvements in our results, showcasing a direct correlation between the number of paraphrases utilized and the models' accuracy. Our experimentation revealed a compelling trend, shown in Figure 4. For all models tested, the accuracy was a monotonically increasing function of the number of paraphrases used. This observation strongly suggests the positive impact of paraphrase augmentation, yielding information repetition, on the model's ability to comprehend and generalize new knowledge from limited data.

Learning New Information In Figure 3, we can see an interesting phenomenon observed throughout our experiments. After each epoch, i.e., completing another iteration over the entire dataset, the training loss drops significantly. This is consistent with what is known about LLMs memorizing the data during training and overfitting (Tirumala et al., 2022).

Our hypothesis is as follows:

> In order to teach pre-trained LLMs **new** knowledge, the knowledge must be repeated in numerous ways.

This is well known for LLM pre-training (Kandpal et al., 2023), and we see in this case that this holds for fine-tuning as well. The rationale for this hypothesis is that mere memorization of sentences does not entail knowledge of their content, as was already shown in (Berglund et al., 2023). By providing the information in numerous forms (like the data augmentation process we used), the various relationships in the data (e.g., $a \implies b$, $b \not\Longrightarrow c$) stand a higher chance of appearing naturally. We believe this can potentially both increase $\mathcal{L}_{\mathcal{M},\mathcal{Q}}$ in general, as well as ameliorate Berglund et al.'s *Reversal Curse*. While promising, this result still warrants further research.

7. Conclusion and Future Work

Large language models possess vast amounts of knowledge on various topics. In this work, we tested their capability to adapt to new knowledge: both specialized and completely unseen. This is among the first studies to compare two prominent approaches in this domain, namely fine-tuning and retrieval augmented generation. While fine-tuning can be useful for many use-cases, we found that RAG is a more reliable choice for knowledge injection.

Some aspects of this work still warrant further research. For example, we focused on unsupervised training as our primary fine-tuning method, as opposed to instruction-tuning or RL-based methods. Researching combinations of various techniques, with diverse auxiliary knowledge bases, may yield improved results. This approach, combined with our hypothesis from Section 6, could further enhance our understanding of knowledge injection via FT.

While we believe that this work further enhances our understanding of knowledge in LLMs, there is a lot more work to be done in this field. Specifically, more research is required regarding the question of knowledge representation in LLMs, especially from a theoretical perspective.

Finally, further efforts are needed to measure knowledge in LLMs. While we employed an empirical approach as described in Equation (2), it is important to explore other definitions and perspectives on knowledge as well, and extend upon this work.

8. Limitations

As in all machine learning applications, the choice of hyperparameters significantly impacts the results. We therefore strongly recommend optimizing all relevant hyperparameters for specific cases.

We have supported our claims by running the experiments on three different models. However, generalization to other LLMs should be tested thoroughly. For example, GPT-4 achieves near perfect accuracy for some MMLU tasks (Nori et al., 2023), and thus further improvement is not applicable.

Finally, while we chose various topics for the knowledge bases, all of our sources came from Wikipedia. Other datasets may yield different results, and must be evaluated carefully.

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